

NPTGA Enterprise Incident Investigation Report

INC-99214 — Operation DarkSiphon

MOCK / SYNTHETIC TEST DATA — Fictional organization, personnel, infrastructure, IP addresses, hashes, metrics and incident relationships. Intended only as test data for SIH26154 document intelligence pipelines. CVE identifiers are included as requested test entities; descriptions here are synthetic and are not authoritative vulnerability advisories.

Page 1 — Document Header & Executive Summary

Organization: National Power & Telemetry Grid Authority (NPTGA)

Classification: TLP:AMBER | RESTRICTED

Incident Identifier: INC-99214 (Operation DarkSiphon)

Date Range: September 2026

On 07 September 2026, NPTGA detected anomalous outbound traffic from a telemetry aggregation segment supporting regional SCADA operations. The investigation reconstructed a multi-stage intrusion attributed, for this synthetic exercise, to APT-44 (VortexPhantom). The activity involved exploitation of an internet-facing edge firewall exposure, unauthorized access to management infrastructure, credential abuse, lateral movement into telemetry-support systems, collection of operational datasets and staged exfiltration. Eighteen SCADA telemetry nodes were confirmed compromised or materially affected. Protective isolation caused a cumulative operational interruption of 5 hours 42 minutes across four regional facilities. Investigators estimate 620 GB of data was staged or exfiltrated and assess combined remediation and economic exposure at approximately \$3,450,000 USD. No evidence in this mock dataset indicates permanent damage to protection-relay firmware; temporary loss of telemetry visibility required conservative operating procedures and manual validation.

Key Quantitative Metrics

Metric	Value	Interpretation
Compromised SCADA telemetry nodes	18	Confirmed affected telemetry endpoints
Data volume exfiltrated	620 GB	Network-flow/storage reconciliation estimate
Total outage / degraded operation	5 h 42 min	Cumulative across four facilities
Estimated remediation + economic exposure	\$3,450,000 USD	Response, replacement, downtime and business exposure
Affected facilities	4	North, Central, East and Coastal
Peak anomalous outbound rate	186 Mbps	Observed during staging/transfer
Credential principals reset	73	Human and service identities
Forensic evidence items	146	Disk, memory, log, network and firmware evidence

Scope & Evidence

• Firewall, VPN/authentication, endpoint, SCADA historian, NetFlow, system-image, firmware and BGP evidence.
• Evidence window: 07–18 September 2026 UTC.
• Repeated entities are intentional for extraction and consistency testing.

Page 2 — Threat Actor Attribution & Technical Attack Vector

Threat Actor: APT-44 (VortexPhantom). Attribution is assessed as HIGH confidence for this synthetic exercise based on infrastructure reuse, authentication behavior, command sequencing, tooling metadata and temporal correlation. Initial access was reconstructed as exploitation of an edge firewall vulnerability represented by CVE-2024-21762, followed by an authentication-bypass condition represented by CVE-2024-3400. The actor established persistence on NPTGA-JMP-07, reused privileged credentials, enumerated the telemetry-support VLAN, collected historian data and staged compressed archives on NPTGA-STG-03. The report intentionally repeats hostnames, IPs, dates, hashes and identifiers to test entity resolution and cross-output consistency.

Technical Timeline (UTC)

UTC Timestamp	Hostname	Source → Destination	Event
2026-09-07 02:14:33	EDGE-FW-02	10.44.0.14 → 198.51.100.27	Suspicious edge-management request
2026-09-07 02:19:08	EDGE-FW-02	10.44.0.14 → 198.51.100.27	Unexpected privileged process
2026-09-07 02:37:51	NPTGA-JMP-07	10.44.18.27 → 10.44.32.11	Administrative session
2026-09-07 03:06:44	NPTGA-JMP-07	10.44.18.27 → 10.44.32.0/24	Credential reuse/lateral enumeration
2026-09-08 11:42:17	SCADA-TLM-14	10.44.32.114 → 10.44.50.9	Abnormal historian queries
2026-09-10 23:16:02	NPTGA-STG-03	10.44.18.53 → 10.44.60.18	84.6 GB archive staging
2026-09-12 04:52:29	NPTGA-STG-03	10.44.18.53 → 198.51.100.41	Outbound transfer begins
2026-09-14 17:08:55	SCADA-TLM-03	10.44.32.103 → 10.44.18.53	Repeated collection
2026-09-17 08:31:12	SOC-SENSOR-01	10.44.90.12 → 10.44.18.53	Detection rule triggered
2026-09-18 13:45:00	EDGE-RTR-01	10.44.0.1 → N/A	Containment initiated

Indicators of Compromise

Indicator Type	Value	File Path	SHA-256 Hash	Action
IPv4	198.51.100.27	N/A	N/A	Block at perimeter
IPv4	198.51.100.41	N/A	N/A	Block and hunt flows
Hostname	NPTGA-JMP-07	N/A	N/A	Isolate and reimagine
File	syncsvc.bin	C:\ProgramData\NPTGA\syncsvc.bin	7f4e0a9c1a3b7d2e8f6c4b1d9a0e5c7f3b2a1d8e6c9f0a4b7d5e2c8f1a6b3d9e	Quarantine
File	telemetry-cache.so	/opt/nptga/lib/telemetry-cache.so	d1b7c4e90a6f2d3c8e1b5a7f4c9d0e2a6b8c3f1d7e5a9b0c4d2f6e8a1b3c7d	Quarantine

Page 3 — Impact Analysis by Facility & Operational Division

Containment prioritized safety and observability over continuity. Operators placed affected telemetry channels into manual-validation procedures while engineering teams isolated the telemetry-support VLAN. Central Grid Operations experienced the longest disruption because three aggregation paths required simultaneous credential rotation and image verification. North and East retained physical generation and transmission control but temporarily lost portions of centralized telemetry visibility. Coastal experienced the largest data-collection anomaly because its historian retained high-resolution operational records. No mock evidence establishes destructive modification of protection-relay logic; the principal operational effect was reduced confidence in telemetry integrity and availability.

Facility Damage Breakdown

Facility Name	Regional Zone	Disrupted Relays	Recovery Time	Direct Economic Impact
North River Switching Complex	North Zone	7	1 h 18 min	\$420,000 USD
Central Load Dispatch Center	Central Zone	12	2 h 31 min	\$1,280,000 USD
East Valley Substation Cluster	East Zone	6	1 h 07 min	\$690,000 USD
Coastal Generation & Telemetry Hub	Coastal Zone	9	46 min	\$560,000 USD

Impact Classification

Domain	Severity	Primary Finding
Confidentiality	High	Operational telemetry/admin datasets collected and exfiltrated
Integrity	Medium	No confirmed destructive relay-logic modification
Availability	High	5 h 42 min cumulative degraded operation/outage
Safety	Low–Medium	Manual validation increased workload; no simulated safety-system failure
Recovery	Medium	18 nodes investigated; 11 required rebuild

Operational Division Findings

• Grid Operations: manual validation and increased operator callouts.
• SCADA Engineering: 18 nodes investigated; 11 rebuilt.
• Identity & Access Management: 73 human/service credentials reset.
• Network Operations: telemetry-support VLAN isolated and selected BGP routes withdrawn.
• Security Operations Center: firewall, endpoint, authentication and NetFlow evidence correlated.

Page 4 — Containment, Eradication & Forensic Evidence

At 13:45 UTC on 18 September 2026, NPTGA initiated emergency isolation. EDGE-FW-02 management exposure was restricted to the security administration enclave. NPTGA-JMP-07 and NPTGA-STG-03 were removed from normal routing and preserved for forensic acquisition. Telemetry-support VLAN 10.44.32.0/24 was placed behind an explicit deny-by-default egress policy, while approved management and time-synchronized monitoring paths remained available. Operators validated critical telemetry against local indicators until affected nodes passed integrity review. Kerberos TGT/TGS tickets were purged for 73 affected principals; two staged KRBTGT rotations were performed; 31 service credentials and 42 privileged credentials were reset; and 26 high-privilege accounts were re-enrolled for MFA.

Credential Response

Control	Action	Result
Kerberos TGT/TGS revocation	Purged active tickets for 73 affected principals	Sessions invalidated
KRBTGT rotation	Two staged key rotations with replication validation	Forged-ticket risk reduced
Service-account rotation	31 service credentials regenerated	Telemetry/historian services restored
Privileged admin rotation	42 privileged credentials reset	Controlled access restored
MFA re-enrollment	26 high-privilege accounts re-enrolled	Additional assurance

Cryptographic Verification

Evidence Class	Quantity	Verification Method	Outcome
Disk images	28	SHA-256 acquisition/recheck	28/28 matched
Memory captures	17	Acquisition hash + chain of custody	17/17 preserved
Log bundles	39	SHA-256 + timestamp correlation	39/39 matched
Network-flow exports	24	Hash + source reconciliation	24/24 matched
Firmware/integrity records	17	Manifest/version comparison	17/17 passed

Exact Containment Firewall Rules

Rule ID	Direction	Source	Destination	Protocol/Port	Action
CF-001	OUT	10.44.32.0/24	ANY	ANY	DENY + LOG
CF-002	OUT	10.44.18.53/32	198.51.100.0/24	TCP/443	DENY + LOG
CF-003	IN	ANY	10.44.18.27/32	TCP/22,3389	DENY + LOG
CF-004	IN	SOC-ADMIN-NET	10.44.0.14/32	TCP/443	ALLOW + LOG
CF-005	OUT	10.44.32.0/24	10.44.90.12/32	UDP/123	ALLOW + LOG

BGP Route Withdrawals: 10.44.32.0/24, 10.44.18.0/24 and 10.44.60.0/24 were withdrawn at EDGE-RTR-01. Routing validation occurred at 13:49, 13:51 and 13:53 UTC.

Page 5 — Strategic Recommendations & Executive Board Mandates

The investigation demonstrates how perimeter compromise can become an operational risk when identity, administrative jump hosts, telemetry-support networks and data repositories share excessive trust relationships. NPTGA should use the incident to reduce implicit trust, make sensitive actions attributable, preserve immutable evidence of administrative and transformation events, and ensure future investigations can reconstruct source-to-output provenance without exposing sensitive source material to uncontrolled systems.

Five Prioritized Remediation Mandates

1. Zero Trust Segmentation — CRITICAL: identity-aware access across IT, OT/SCADA, telemetry support and administration. Target 31 March 2027. Allocation \$1,200,000 USD.
2. Privileged Access Modernization — CRITICAL: hardened jump hosts, phishing-resistant MFA, just-in-time privilege, vaulting and service-account rotation. Target 31 January 2027. Allocation \$650,000 USD.
3. Immutable Provenance Logging — HIGH: append-only audit records and permissioned-ledger anchoring of finalized artifact hashes/minimum metadata. Allocation \$400,000 USD.
4. Telemetry Integrity Program — HIGH: cryptographic verification of system images, firmware manifests and configuration baselines. Allocation \$525,000 USD.
5. Continuous Detection & Recovery Exercise — MEDIUM: expand NetFlow analytics and quarterly recovery exercises; target simulated reconstruction within 60 minutes. Allocation \$325,000 USD.

Budget Allocation

Program	Allocation	Outcome
Zero Trust architecture	\$1,200,000 USD	Segmentation and identity-aware access
Privileged access modernization	\$650,000 USD	Reduced credential/lateral-movement risk
Immutable provenance logging	\$400,000 USD	Tamper-evident audit/artifact history
Telemetry integrity	\$525,000 USD	Verified system/firmware baselines
Detection & recovery exercises	\$325,000 USD	Faster detection and resilient operations
TOTAL	\$3,100,000 USD	Two-year security modernization envelope

Sign-Off

Role	Name	Date	Decision
Lead Forensic Investigator	Dr. Meera Khanna	18 September 2026	Findings accepted for remediation tracking
Chief Information Security Officer (CISO)	Arjun Malhotra	19 September 2026	Remediation mandates approved for board submission